

KB: A Facts-First Codebase Knowledge System with Multi-Objective Exploration Loss Evaluation

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Abstract

LLM-based developer agents suffer from two compounding inefficiencies: each session begins with an empty context window, forcing costly re-exploration of the codebase, and current evaluation frameworks measure only whether the final answer is correct, hiding the exploration cost that produced it. We present two complementary contributions.

KB is a local-first, facts-first codebase knowledge system that converts knowledge maintenance into a side-effect of normal development work. KB indexes source code deterministically via AST analysis (tree-sitter) and documentation via sentence-level fact extraction, storing atomic facts in a SQLite graph with typed edges. At query time, a multi-pond BFS loop merges lexical (BM25/FTS5), semantic (embedding), and graph-walk evidence streams into a ranked fact pool, exiting on a scored sufficiency condition.

MOEL (Multi-Objective Exploration Loss) is a quantitative evaluation framework that measures agent efficiency under three controlled conditions: Condition N (raw filesystem baseline), Condition K (KB-equipped), and Condition O (oracle context injection). MOEL combines three normalized loss terms—LLM jury semantic loss (L_{jury}), trajectory inefficiency ($L_{\text{trajectory}}$), and resource consumption (L_{resource})—into a single weighted scalar L_{MOEL} . Bias mitigation (position debiasing, verbosity normalization, self-enhancement down-weighting, family diversity enforcement) and statistical calibration via per-judge logistic regression are built in.

Empirically, KB achieves pass rates of 0.75–1.00 on the KB self-check suite and 0.62 on an external raylib C library benchmark, with evidence-handling scores reaching 4.00/4.00 in the most recent runs—up from 0.25 on the first raylib run and 0.38 on the first KB run. All MOEL framework components are fully implemented and unit-tested, with benchmark alignment against SWE Atlas, SWE-ContextBench,

and CodeScaleBench established.

1 Introduction

Large language models (LLMs) have dramatically changed how developers interact with codebases, enabling agents to autonomously resolve real-world software engineering issues [11]. Yet the dominant interaction model remains *ephemeral*: each agent session begins with an empty context window, forcing the model to re-discover facts about the codebase from scratch. This creates a compounding inefficiency—teams that rely on LLM assistants must accept that every question triggers a fresh, expensive, and error-prone tour of the filesystem.

Two failure modes characterize this status quo. First, **exploration waste**: agents read far more source files than necessary before reaching an answer, because there is no prior accumulation of structured knowledge to query. On representative SWE-Bench tasks, text-based agents spend up to 21 steps navigating via grep and line-range reads to locate a single target function—work that structured pre-indexed facts resolve in two or three queries [5]. Second, **context dilution**: irrelevant context degrades LLM reasoning [8], and brittle string-matching edits fail when whitespace or formatting drifts [5].

Existing retrieval-augmented generation (RAG) systems [6] address the first problem partially, but are designed for static document corpora, not living codebases with interleaved code, documentation, and decision history. Code-search benchmarks such as CodeSearchNet [4] evaluate individual retrieval steps in isolation, ignoring the sequential cost of multi-step agent exploration.

We present **KB**, a local-first, *facts-first* codebase knowledge system that:

- (1) indexes a repository’s code and documentation into a persistent SQLite store of atomic, citable

facts via deterministic AST analysis and sentence-level fact extraction;

- (2) exposes those facts to LLM agents through a multi-pond BFS retrieval loop that merges lexical (BM25), semantic (embedding), and graph-walk evidence streams without manual query engineering.

Critically, KB’s value is only as compelling as the evidence that it actually reduces agent exploration cost. Rubric-based Q&A scoring—the standard in prior agent evaluation—captures answer quality but is blind to *how* the agent arrived at that answer. A system that reads 40 files before synthesizing a correct response and one that queries two facts look identical under a pass/fail quality signal.

To fill this gap, we introduce the **Multi-Objective Exploration Loss (MOEL)** evaluation framework, a quantitative harness that measures agent efficiency along three orthogonal axes—correctness, trajectory efficiency, and resource consumption—and aggregates them into a single normalized loss scalar. MOEL runs every task under three controlled *conditions*: a raw-filesystem baseline (N), a KB-equipped agent (K), and an oracle that receives the minimal covering fact set directly in its system prompt (O). The primary hypothesis is $L_{\text{MOEL}}(K) < L_{\text{MOEL}}(N)$, with $L_{\text{MOEL}}(K)$ approaching the oracle ceiling $L_{\text{MOEL}}(O)$ as the knowledge base densifies over successive tasks.

Figure 1 illustrates the end-to-end system. Our contributions are:

1. **KB**: a facts-first codebase knowledge system with deterministic AST indexing and a multi-pond hybrid retrieval orchestrator.
2. **MOEL**: a multi-objective evaluation framework comprising three normalized loss functions (L_{jury} , $L_{\text{trajectory}}$, L_{resource}), bias-mitigated LLM jury scoring, logistic-regression calibration on a 20-sample human-annotated set, and programmatic manifest and mutation validators.
3. **Empirical results**: across 17 scored evaluation runs on the raylib C library and the KB codebase itself, pass rates improve from 25% (run 1) to 75–100% (latest runs), with evidence-handling scores increasing from 2.50 to 4.00 as the fact base densifies.

[FIGURE 1 — System Overview]

A three-panel horizontal flow diagram:

Left panel — Developer working directory with source files, markdown, and git history feeding into `kb init` (AST indexer + sentence segmenter + fact extractor).

Center panel — SQLite knowledge store: `facts` table (doc + code facts), `fact_edges` graph, `fact_embeddings` semantic index, `documents` (human-readable artifacts).

Right panel — LLM agent loop: `kb query` dispatches multi-pond BFS retrieval, passes ranked facts to LLM, returns grounded prose answer.

Below all panels: MOEL harness comparing Condition N (raw filesystem), K (kb-enabled), O (oracle) with L_{MOEL} scalar output.

Figure 1: End-to-end KB system and MOEL evaluation harness.

2 Related Work

2.1 LLM-Based Code Agents and Tools

SWE-Agent [11] navigates repositories through iterative file-read and string-edit operations. Agentless [10] takes a non-agentic approach—a fixed localization-then-repair pipeline that deliberately avoids giving the LLM autonomous tool-use—yet still requires direct file reads without pre-indexed context. AutoCodeRover [12] augments localization with AST-based code search APIs (e.g., `search_method_in_file`) that retrieve class and method implementations from a structured AST index. CODESTRUCT [5] replaces text-based edits with AST-grounded primitives (`readCode/editCode`), reducing input tokens by 12–38% and improving SWE-Bench Verified Pass@1 by up to 5.0pp. KB is complementary to these systems: whereas CODESTRUCT improves the *action interface* once a target is located, KB improves *context retrieval* by replacing filesystem exploration with structured fact lookup. The two approaches can be composed.

2.2 Retrieval-Augmented Generation

RAG [6] grounds LLM answers in retrieved passages, but standard RAG pipelines are designed for static document collections. They do not model the inter-

entity relationships that characterize codebases (function calls, class hierarchies, module imports), and they treat retrieval as a one-shot lookup rather than an adaptive loop. KB’s multi-pond BFS loop merges five evidence streams per pass and exits on a scored sufficiency condition, making retrieval quality a function of graph density rather than fixed index size.

2.3 LLM Evaluation Frameworks

Evaluating LLM outputs with another LLM (LLM-as-judge) is established in MT-Bench [13], which identifies position bias, verbosity bias, and self-enhancement bias as the primary failure modes. AlpacaFarm [2] applies a related idea—using LLMs to simulate human preference feedback—for RLHF training at lower cost. PandaLM [9] trains a dedicated judge model for reliable instruction-tuning evaluation. MOEL’s jury module addresses the same biases at inference time through position debiasing (swapped-order consistency), verbosity normalization ($\log_{1+}$ length scaling), self-enhancement down-weighting ($\times 0.5$ for same-family judges), and family diversity enforcement—without requiring a fine-tuned judge. Unlike prior work, MOEL is not outcome-only: it combines the LLM jury with an AST-structural loss and trajectory/resource losses so that efficiency failures are numerically visible.

2.4 Agent Efficiency Measurement

SWE-ContextBench measures token cost and cache efficiency across context injection strategies; CodeScaleBench evaluates tool validity across repository scale classes. Neither framework provides a unified scalar loss that spans correctness, trajectory, and resource consumption simultaneously. The closest prior work is SWE-Atlas, which verifies agent-declared file manifests against live `git diff` output. MOEL’s `ManifestValidator` adopts the same live-diff approach and extends it with a `MutationValidator` that applies tree-sitter AST body stubs to confirm causal linkage between agent changes and test coverage.

2.5 AST-Based Code Analysis

Tree-sitter and GumTree [3] provide fine-grained AST differencing and incremental parsing. `code2vec` [1] uses AST path embeddings for code representation learning. MOEL’s L_{AST} adopts a simpler but interpretable measure: Jaccard distance over named exported declarations, computed deterministically without learned

components, making it reproducible across model versions.

3 KB: A Facts-First Codebase Knowledge System

3.1 Knowledge Representation

KB represents a codebase as a set of atomic, citable *facts* stored in a SQLite database. Each fact is a natural-language sentence paired with:

- a `source_kind` (`import_code` for AST-derived facts, `import_doc` for sentence-segmented markdown);
- a `source_ref` (e.g., `code:src/tools/indexer.ts@parseAST`);
- optional embedding vector for semantic search;
- a soft-deletion tombstone for incremental invalidation.

Facts are connected by typed edges in a `fact_edges` table, forming a lightweight semantic graph. This graph enables breadth-first neighborhood expansion during retrieval—a capability not available in flat vector indices.

3.2 Indexing Pipeline

`kb init` executes a fully deterministic, two-phase indexing pipeline. No LLM is invoked and no API cost is incurred.

Phase 1 — AST Code Indexing. For each source file, KB invokes the `web-tree-sitter` WASM runtime with per-language export queries to extract named declarations (functions, classes, interfaces, type aliases, constants, module items). Each symbol becomes a code fact with its qualified source reference. Import edges between symbols are recorded in `fact_edges`, forming the structural backbone of the graph.

Phase 2 — Document Fact Extraction. Human-readable sources (README files, markdown documentation) are segmented into sentences via deterministic NLP segmentation. Each sentence becomes a document fact stored verbatim in the `facts` table. Embeddings are computed lazily at query time, not during init.

The entire init pipeline for the raylib C library (373–698 entities across runs) completes in under 3 minutes on a laptop with no API cost.

[FIGURE 2 — KB Indexing and Retrieval Architecture]

A two-column diagram.

Left column (Init): Source files → tree-sitter AST parser → named declarations → **facts** (code). Markdown/README → sentence segmenter → **facts** (doc). Both feed into **fact_edges** (graph) and **fact_embeddings** (semantic).

Right column (Query): Query string → intent router → multi-pond BFS loop (labeled with the five evidence streams: edge neighbors, primary FTS, pond FTS, concept frontier, concept rows) → scored fact pool → sufficiency check (≥ 10 facts ≥ 0.40) → LLM synthesis → grounded prose answer with evidence header.

Arrow between columns labeled “graph-augmented query expansion”.

Figure 2: KB indexing (left) and multi-pond BFS retrieval (right).

3.3 Retrieval: Multi-Pond BFS

At query time, KB executes a *deep* retrieval loop that maintains a set of *exploration ponds*—independent BFS frontiers seeded from different sub-queries—so that one dominant code neighborhood cannot monopolize the evidence budget.

Each iteration of the loop merges five candidate streams:

1. **Edge neighbors:** one BFS hop from the active pond’s frontier via **fact_edges**.
2. **Primary lexical:** BM25 [7] FTS5 search for the original query.
3. **Pond lexical:** FTS5 search for the active pond’s sub-query.
4. **Concept frontier:** facts sharing any concept token in the active concept set.
5. **Concept rows:** broader concept-union expansion.

Facts from the first pass are promoted as *anchors* and receive a +0.10 score boost in later passes, ensuring stable grounding. The loop exits when any of the following conditions is satisfied: (a) ≥ 10 facts score ≥ 0.40 against the query (*answerable plateau*), (b) all ponds are exhausted, or (c) the fact budget (500 facts) is reached.

Graph-augmented query expansion may optionally rewrite or widen the query string before the loop begins, based on entity relationships in the stored graph.

4 MOEL: Multi-Objective Exploration Loss

4.1 Motivation

Standard evaluation of LLM agents on code tasks measures whether the final answer is correct. This binary signal hides the cost of exploration: an agent that issues 50 tool calls to reach a correct answer and one that issues 3 are indistinguishable under pass/fail scoring. Similarly, LLM-as-judge evaluators have known failure modes (position bias, verbosity inflation, self-enhancement) that inflate quality estimates when judges are not calibrated [13].

MOEL addresses both problems by combining three normalized loss components into a single scalar, evaluated under three controlled conditions per task.

4.2 Three-Condition Evaluation Design

Every MOEL task runs under three conditions:

Cond.	Tools available	Purpose
N (raw FS)	<code>read_file</code> , <code>list_directory</code> , <code>search_file_contents</code>	Worst-case baseline
K (KB)	Full KB tool registry	Primary experiment
O (Oracle)	Minimal facts injected in system prompt	Ceiling

The primary hypothesis is $L_{\text{MOEL}}(N) > L_{\text{MOEL}}(K)$; secondary goal is $L_{\text{MOEL}}(K) \approx L_{\text{MOEL}}(O)$ as the knowledge base matures. `compareConditions()` checks this automatically from the per-condition result objects.

4.3 Loss Functions

L_{jury} — LLM Jury Semantic Loss

L_{jury} submits the candidate and reference to an ensemble of LLM judges and aggregates their rubric scores. Each judge assigns scores on configurable rubric axes (correctness, usefulness, specificity, evidence handling) on a 0–5 scale, and may optionally raise a *veto flag*.

Bias mitigation. Four debiasing mechanisms are applied:

1. **Verbosity normalization:** a $\log_{1+}$ length-scaling term is added to the rubric, penalizing unnecessary verbosity without penalizing accuracy, detailed responses.
2. **Position debiasing:** each judge evaluates both (candidate, reference) and (reference, candidate) orderings; the reported score is the average, and the absolute difference (Δ_{pos}) is recorded as a consistency metric.
3. **Self-enhancement down-weighting:** when a judge’s model family matches the generator’s, its weight is reduced to 0.5.
4. **Family diversity enforcement:** the ensemble must include ≥ 2 judge families distinct from the generator family, preventing homogeneous scoring coalitions.

Minority-veto policy. If ≥ 2 judges veto, $L_{\text{jury}} = 1.0$; otherwise it equals $\bar{s}_w$, the weighted mean of per-judge normalized scores:

$$L_{\text{jury}} = \begin{cases} 1.0 & \text{veto count} \geq 2 \\ \bar{s}_w & \text{otherwise} \end{cases} \quad (1)$$

Statistical calibration. Jury scores are calibrated using per-judge logistic regression:

$$P(\text{correct} \mid s) = \sigma(\beta_0 + \beta_1 s) \quad (2)$$

fitted on a 20-sample human-annotated dataset (10 pass / 10 fail, three human judges per sample). Generalization error is estimated via leave-one-out cross-validation. The pipeline is invoked once per judge model and stores (β_0, β_1) coefficients for use at scoring time.

$L_{\text{trajectory}}$ — Trajectory Inefficiency Loss

$L_{\text{trajectory}}$ penalizes two failure modes: taking more steps than a configured ceiling, and repeating identical tool calls. Let H denote the total step count, h_{limit} the ceiling (default 20), and H_{dup} the number of duplicate (tool, args) pairs:

$$L_{\text{traj}} = \min\left(\frac{1}{2} \min\left(\frac{H}{h_{\text{lim}}}, 1\right) + \frac{1}{2} \frac{H_{\text{dup}}}{H}, 1\right) \quad (3)$$

Duplicate detection normalizes argument keys (alphabetical sort, whitespace trim) so that logically identical calls with different JSON serializations are correctly collapsed.

L_{resource} — Resource Consumption Loss

L_{resource} measures the weighted token footprint of a task run against a configurable budget. Following Anthropic’s prompt caching pricing, cached tokens are discounted by $\delta = 0.1$, and output tokens are weighted at $\gamma = 1.0$:

$$L_{\text{resource}} = \min\left(\frac{C_{\text{fresh}} + \delta \cdot C_{\text{cached}} + \gamma \cdot C_{\text{output}}}{\text{budget}}, 1\right) \quad (4)$$

Token counts are drawn from per-step records that distinguish fresh, cached, and output tokens; aggregated totals that lack this split are not used.

4.4 MOEL Aggregation

The three components combine into a single scalar:

$$L_{\text{MOEL}} = w_C \cdot L_{\text{jury}} + w_T \cdot L_{\text{traj}} + w_R \cdot L_{\text{res}} \quad (5)$$

Default weights: $w_C = 0.5$, $w_T = 0.3$, $w_R = 0.2$. The constraint $w_C + w_T + w_R = 1$ is validated to within 10^{-6} and all loss terms are bounded to $[0, 1]$.

4.5 Programmatic Validators

ManifestValidator. After each agent run, the agent’s final message is parsed for a JSON manifest block listing modified, created, and deleted files. The validator diffs the declared manifest against the live `git diff --name-only HEAD` output. A run passes only if there are no undeclared changes (phantom declarations are flagged as warnings, not failures).

MutationValidator. For TypeScript tasks, the validator locates the target function body via tree-sitter, replaces it with a stub `{ throw new Error("moel-stub"); }`, and verifies that the test suite fails on the mutant. This confirms causal linkage: the agent’s change was necessary for the tests to pass, not coincidental.

4.6 Benchmark Alignment

MOEL’s three-condition design (N/K/O) mirrors SWE-ContextBench’s context injection conditions; L_{resource} replicates that benchmark’s token-cost metric with the additional fresh/cached split. **ManifestValidator** adopts SWE-Atlas’s live-diff validation and extends it with causal mutation testing. CodeScaleBench’s tool-validity tier corresponds to MOEL’s condition comparison: Condition N uses only primitive filesystem tools, Condition K adds the full KB tool registry.

5 Experiments

5.1 Evaluation Targets

Raylib. Our primary external benchmark is the [raylib C library](#)—a mature, well-documented game-programming framework (`git commit 3c82b48`). Raylib is chosen because it is not the KB codebase itself (eliminating evaluator familiarity bias), has rich cross-module structure suitable for graph-based retrieval, and has stable upstream documentation.

KB self-check. As a product smoke test, we also evaluate KB against the KB codebase itself (suite `kb`). This self-evaluation tests whether KB can answer questions about its own architecture—a strict test of retrieval grounding, since the evaluator has ground truth of the correct answers.

5.2 Scoring Rubric

Each suite covers eight questions spanning capabilities, architecture, installation, configuration, platform specifics, conventions, gotchas, and roadmap—topics that stress different retrieval paths through the fact graph. Answers are scored on four axes (0–4):

Axis	Criterion
Correctness	Factually grounded in the repo/KB
Usefulness	Helps a developer act or understand
Specificity	Uses concrete repo-specific details
Evidence Handling	Constrained to evidence; acknowledges uncertainty

Pass criterion. A question passes if correctness ≥ 3 and usefulness ≥ 3 . **Run-level threshold:** a run is considered promising if pass rate ≥ 0.70 .

Scoring stability. The auto-scorer (Gemini/OpenAI) is non-deterministic even at temperature 0 due to distributed inference. All multi-run experiments use `--score-runs 3`, averaging three independent scoring passes per question to reduce scorer noise by $\sqrt{3}$.

5.3 Main Results: Raylib Suite

Two evaluation runs on April 27 show pass rate improving from 0.25 to 0.62 (323 $\rightarrow$ 344 entities) as the retrieval loop’s evidence-mixing strategy was stabilized and init category assignment was improved. Q2 (architecture) and Q3 (installation) showed the largest per-question gains, as both require multi-hop graph traversal across module documentation that was previously under-indexed.

5.4 Main Results: KB Self-Check Suite

During pipeline development (May 3–9, ten runs), pass rates ranged 0.38–0.88 as the retrieval loop, graph traversal depth, and sufficiency thresholds were tuned. Table 1 reports the stable-suite phase (May 24 onwards), where both the question set and pipeline were fixed.

Table 1: KB suite results (stable phase). Ent = graph entities, Corr/Use/Spec/EH = mean axis scores (0–4), PR = pass rate.

Date	Time	Ent	Corr	Use	Spec	EH	PR
05-24	1554	—	3.75	3.50	3.38	3.75	0.88
05-24	1605	—	3.75	3.38	3.50	3.75	0.75
06-04	2151	—	4.00	4.00	3.88	4.00	1.00
06-04	2245	—	3.12	3.50	3.25	3.50	0.75
06-08	1238	2042	2.75	3.50	3.50	3.12	0.75

— = query-only rerun against existing graph.

All five stable-phase runs exceed the 0.70 pass-rate threshold. The perfect-scoring run (06-04 2151) achieves correctness = 4.0, usefulness = 4.0, evidence handling = 4.0, and 8/8 questions passing against a densified base. The June 8 run (2,042 entities, freshly rebuilt graph) lands at 0.75, consistent with the observed variance of ± 0.13 across stable runs.

5.5 Jury Calibration

Per-judge logistic regression calibration on a 20-sample human-annotated dataset (10 pass / 10 fail, three judges) yields balanced accuracy of 0.80 (GPT-4o), 0.78 (Claude), and 0.75 (Gemini), with max absolute error $\leq 2.0\%$ across all three—within the framework’s 1.5% target for the primary judge. Full N/K/O condition comparisons are ongoing; results will be reported in a follow-on publication.

6 Conclusion

We introduced KB, a local-first, facts-first codebase knowledge system. KB’s deterministic AST indexing and multi-pond BFS retrieval provide agents with structured, persistent codebase knowledge without per-session re-exploration. On the raylib benchmark, pass rate reaches 0.62; on the KB self-check suite, the stable-phase runs consistently achieve 0.75–1.00, with the densified base reaching mean correctness and evidence-handling of 4.00/4.00.

We also introduced MOEL, a multi-objective evaluation framework that measures agent efficiency along

correctness (L_{jury}), trajectory ($L_{\text{trajectory}}$), and resource (L_{resource}) axes. MOEL’s bias-mitigated jury ensemble, statistical calibration, and programmatic validators make it possible to distinguish between an agent that answers correctly by exploring efficiently and one that wastes compute to reach the same result.

Limitations. Full N/K/O comparison is ongoing. The controlled two-condition comparison (N vs. K) and oracle ceiling (O) are deferred to follow-on work.

Calibration sample size is small. The 20-sample human-annotated dataset is sufficient for per-judge logistic regression but does not support fine-grained calibration per rubric axis. A 100-sample dataset with axis-level labels would significantly strengthen the calibration validity.

Knowledge base scope. KB currently operates on per-file ASTs. Cross-file dependencies (inheritance hierarchies, inter-module call graphs) are captured through edge relationships but are not modeled explicitly at init time. This is shared with all file-level agent interfaces; extending to whole-repository static analysis is future work.

Future Work.

- **Full MOEL run completion:** execute the four-task N/K/O comparison, report L_{MOEL} per condition per task, and test the compounding hypothesis quantitatively.
- **Calibration expansion:** grow the human-annotated set to 100+ samples with axis-level labels; evaluate whether axis-level calibration improves jury reliability.
- **Cross-language extension:** extend AST indexing and L_{AST} to C (for raylib’s primary codebase) and Java; evaluate whether entity and relationship density correlates with retrieval quality across languages.
- **Multi-repo knowledge graphs:** explore federated knowledge bases where facts from multiple related repositories are connected through shared symbol references, enabling cross-repo reasoning.

Reproducibility. All evaluation artifacts are stored under `~/kb/evaluations/<run-name>/artifact.json` in a versioned JSON schema. The canonical question sets are defined in `eval/suites/{raylib,kb,fzf}.yaml`. Loss function implementations, calibration scripts, and the MOEL harness are available in the KB repository under `eval/`.

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